DS3040 Introduction to Deep Learning Lab

Assignment-3, Date: Jan 27, 2025

Timing: 10:00 AM to 12:45 PM   Max Marks: 20

Instructions

1. Submit one .ipynb file containing all answers. The name should be [student_name] _assignment[number].ipynb
2. Write the questions in separate text blocks before the answers.
3. Outputs for all sub-questions should be given and the code should be executable.
4. Write justifications for your choices where needed.
5. Ensure that all plots include clear labels and legends for better interpretation.
6. Do not use AI or any websites except library documentations.

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1. Implement a Feedforward Neural Network (FNN) using the TensorFlow/Pytorch library. (5)

   a Load the Iris dataset from *sklearn.datasets*. Preprocess and standardize the features.

   b Build an FNN with at least one hidden layer using the ReLU activation function.

   c Train the model and Report the training and validation accuracy/loss.

   d Evaluate the model on the test set and provide the confusion matrix, test accuracy, and classification report.

2. Experiment with hyperparameter tuning to optimize your FNN. (5)

   a Modify the following hyperparameters:
      - Number of hidden layers or neurons.
      - Learning rates.
      - Batch size.
      - Dropout layers
      - Normalization

   b Train the updated model and compare its accuracy with the original model.

   c Provide updated plots for training and validation accuracy/loss, and evaluate the new model on the test data.

   d Discuss the impact of your changes and identify the best-performing configuration.

3. Implement a Feedforward Neural Network (FNN) for a regression problem using the Boston Housing dataset. (5)

a Load the Boston Housing dataset from *sklearn.datasets* and preprocess the features.

b Build an FNN with at least one hidden layer.

c Train the model and Report the training and validation loss.

d Evaluate the model on the test set and provide the test loss and other relevant regression metrics(RMSE, R-squared).

4. Implement and train your FNN on the MNIST dataset. (5)

a Load the MNIST dataset using *tensorflow.keras.datasets* . Preprocess the dataset.

b Build an FNN and Train the model on the MNIST dataset. Report the training and validation accuracy/loss.

c Evaluate the model on the test set and provide the confusion matrix, test accuracy, and a brief discussion of the results.

d Compare the results with those obtained from the Iris dataset. Highlight key differences in performance and any challenges encountered due to the increased complexity of the MNIST dataset.